

EE-UY/CS-UY 4563: Introduction to Machine Learning

Midterm 1, Fall 2018

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Answer all THREE questions. Exam is closed book. No electronic aids. But, you are permitted a limited number of cheat sheets. Part marks are given. If you do not remember a particular python command or its syntax, use pseudo-code and state what syntax you are assuming.

Best of luck!

1. Suppose we wish to fit a function of the form,

$$\hat{y} = f(x_1, x_2) = c_0 + \frac{c_1 x_1 (1 + c_2 x_2)}{1 + x_1},$$

where $c = (c_0, c_1, c_2)$ are model parameters.

- (a) Write $f(x_1, x_2)$ as a linear model,

$$f(x_1, x_2) = \sum_{j=0}^{p-1} \phi_j(x_1, x_2) \beta_j,$$

for parameters β_j and basis functions $\phi_j(x_1, x_2)$. State p , write the parameters $\beta = (\beta_0, \dots, \beta_{p-1})$ in terms of the original parameters $c = (c_0, c_1, c_2)$ and write the expressions for the basis functions $\phi_j(x_1, x_2)$.

- (b) Given parameters $\beta = (\beta_0, \dots, \beta_{p-1})$ in the linear model in part (a), what are the corresponding original parameters $c = (c_0, c_1, c_2)$. That is, write c in terms of β .
- (c) Given training data (x_i, y_i) , $x_i = (x_{i1}, x_{i2})$, $i = 1, \dots, n$, find the derivative of the RSS with respect to β_k :

$$\frac{\partial \text{RSS}}{\partial \beta_k} \text{ for } \text{RSS}(\beta) = \sum_{i=1}^n (y_i - \hat{y}_i)^2.$$

You may leave your answer in terms of the functions $\phi_j(\cdot)$ and parameters β_j . You do not need to evaluate the sum.

- (d) Write a python function,

```
c = fit(X,y)
```

that finds the parameters c in your model using training data $\mathbf{x}$ and $\mathbf{y}$ for n samples. You may use the following two functions:

- `beta = lstsq(A,b)`: Solves $\beta = (A^T A)^{-1} A^T b$.
- `A = np.column_stack((col1,...,colp))`: Creates a matrix $\mathbf{A}$ with columns `col1, ..., colp`.

2. Consider a model of the form,

$$\hat{y} = f(x, \beta) = \beta_0 + \beta_1 x + \beta_2 \max\{0, x - 1\}, \quad (1)$$

where $\beta = (\beta_0, \beta_1, \beta_2)$ are parameters.

- (a) Show that the function $2|x - 1|$ is in the model class (1). That is, find the parameter vector β such that $f(x, \beta) = 2|x - 1|$ for all x .
- (b) Now suppose that the true relation between x and y is,

$$y = f_0(x) = \min\{2|x - 1|, 4\}.$$

A model of the form (1) is fit using 100 training samples (x_i, y_i) with $x_i \in [0, 2]$ and $y_i = f_0(x_i)$. What is the parameter vector $\hat{\beta}$ that would be obtained? Explain. Make any reasonable assumptions.

- (c) Draw $f_0(x)$ and the estimate $f(x, \hat{\beta})$ from part (b) for $x \in [0, 4]$. What is the bias,

$$\text{Bias}(x, \hat{\beta}) = f_0(x) - f(x, \hat{\beta}),$$

at $x = 4$?

- (d) There are two models – model 0 and model 1 – with model 0 being simpler. You perform K -fold cross validation and obtain a python array `RSS` where

$$\text{RSS}[i, k] = \text{test RSS for model } i=0, 1 \text{ in fold } k=0, 1, \dots, K-1.$$

Write python code to select the best model (either model 0 or 1) using:

- (i) The normal rule (select the model with the lower mean test RSS)
- (ii) The one SE rule (select the simplest model within one SE of the lowest mean test RSS).

3. You are given training data with features $x_i = (x_{i1}, x_{i2})$ and a binary class label $y_i \in \{0, 1\}$.

i	x_{i1}	x_{i2}	y_i
1	1	0	1
2	-1	0.5	1
3	1	2	0
4	0	-3	0
$\vdots$	$\vdots$	$\vdots$	$\vdots$

Table 1: Training data for the model. Only the first four samples are shown.

- (a) Draw the four points on a scatter plot with different markers for the two classes.
 (b) Consider a classifier of the form,

$$\hat{y} = \begin{cases} 1 & \text{if } z > 0, \\ 0 & \text{if } z < 0 \end{cases} \quad z = \beta_0 + \beta_1 x_1 + \beta_2 x_1^2 + \beta_3 x_2 + \beta_4 x_2^2.$$

Find parameters $\beta = (\beta_0, \dots, \beta_4)$ that make a minimum number of errors on the four samples in Table 1.

- (c) You are now given a classifier of the form,

$$P(y = 1|x) = \frac{1}{1 + e^{-z}}, \quad z = \beta_0 + \beta_1 x_1 + \beta_2 x_1^2 + \beta_3 x_2 + \beta_4 x_2^2.$$

What value of $x = (x_1, x_2)$ maximizes $P(y = 1|x)$ for

- (i) $\beta = (1, 2, -3, 4, -5)$.
 (ii) $\beta = (1, 2, 3, 4, 5)$.
 (d) Finish the following python function that takes a data matrix **x**, vector **y** and parameters **beta** and returns a vector **reject** where **reject[i]=1** if $P(y_i|x_i) < t$.

```
def outlier_detect(X,y,beta,t):
    ...
    return reject
```